

B.Tech/ M.Tech Integrated DEGREE EXAMINATION, MAY 2024
First and Second Semester

21PYB102J – SEMICONDUCTOR PHYSICS AND COMPUTATIONAL METHODS

(For the candidates admitted from the academic year 2022-2023 onwards)

Note:

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

PART – A (20 × 1 = 20 Marks)

Marks BL CO PO

Answer **ALL** Questions

- | | | | | |
|---|---|---|---|---|
| 1. According to Quantum free electron theory, the free electrons move with _____ potential. | 1 | 1 | 1 | 1 |
| (A) Zero | | | | |
| (B) Constant | | | | |
| (C) Infinite | | | | |
| (D) Different | | | | |
| 2. As stated in band theory of solids, which of the following has a zero band gap. | 1 | 1 | 1 | 1 |
| (A) Metal | | | | |
| (B) Semi conductor | | | | |
| (C) Insulator | | | | |
| (D) Both insulator and semi conductor | | | | |
| 3. The range of wave number (k) for the first Brillouin zone is _____ | 1 | 1 | 1 | 1 |
| (A) $-\pi/a$ to π/a | | | | |
| (B) 0 to π/a | | | | |
| (C) $-2\pi/a$ to $2\pi/a$ | | | | |
| (D) 0 to $2\pi/a$ | | | | |
| 4. Which of the following method uses primitive cell approach to compute electronic band structure computationally? | 1 | 2 | 1 | 1 |
| (A) Weigher – seitz method | | | | |
| (B) APW method | | | | |
| (C) Pseudopotential method | | | | |
| (D) Tight bonding method | | | | |
| 5. The fermi level in a N-type semi conductor _____ with increase in temperature. | 1 | 2 | 2 | 1 |
| (A) Becomes zero | | | | |
| (B) Decreases | | | | |
| (C) Remain unchanged | | | | |
| (D) Increases | | | | |
| 6. For a PN junction diode, the current in reverse bias may be _____ | 1 | 1 | 2 | 1 |
| (A) Few milliamperes | | | | |
| (B) Several amperes | | | | |
| (C) Few microamperes to nanoamperes | | | | |
| (D) Few milliamperes to several amperes | | | | |
| 7. Which process of the electron hole pair is responsible for emitting of light? | 1 | 1 | 2 | 1 |
| (A) Generation | | | | |
| (B) Ionisation | | | | |
| (C) Recombination | | | | |
| (D) Diffusion | | | | |

8. Photodiode operate on _____ 1 1 2 1
 (A) Reverse bias (B) Forward bias
 (C) Without bias (D) Ohmic junction
9. Optical processes directly involve _____ absorption and emission 1 2 3 3
 (A) Photon (B) Electron
 (C) Hole (D) Neutron
10. The generation of electron – hole pairs by the incidence of light followed by their radiative recombination is known as 1 2 3 3
 (A) Photoluminescence (B) Electroluminescence
 (C) Thermoluminescence (D) Cathodoluminescence
11. The Fermi's golden rule helps to identify the 1 1 3 3
 (A) Momentum of electrons (B) Velocity of light
 (C) Absorption energy of electron (D) Transition rate per unit volume
12. The optical joint density of states is proportional to 1 1 3 3
 (A) The square root of the energy (B) Square of the energy
 (C) Independent of energy (D) Cube of the energy
13. Two probe technique is suitable for measuring electrical resistivity of _____ samples. 1 1 4 1
 (A) Low resistivity (B) High resistivity
 (C) Magnetic (D) Biological
14. A _____ is a method of determining quickly whether a semiconductor sample is n type (or) p type. 1 1 4 1
 (A) Two point probe (B) Four point probe
 (C) Hot point probe (D) Capacitance – voltage
15. The resistance of a material is inversely proportional to the _____ 1 2 4 1
 (A) Length (B) Square of length
 (C) Area of cross section (D) Area $\times$ length
16. TCAD is a computer simulation technique that is widely used in the semiconductor industry. TCAD is an acronym for 1 1 4 1
 (A) Technology Computer Aided Design (B) Technology Computer Aided Development
 (C) Technology Computer Advances Design (D) Technology Computer Applications Design
17. Nanostructures have sizes in between _____ 1 1 5 3
 (A) 1 and 100 Å (B) 1 and 100 nm
 (C) 100 and 1000 nm (D) 1 and 1000 mm
18. Carbon nanotube reactivity is related to _____ 1 1 5 3
 (A) Volume (B) Length
 (C) Diameter (D) Breadth

PART – C ($1 \times 15 = 15$ Marks)

Marks BL CO PO

Answer ANY ONE Question

- | | | | | | |
|-------|---|----|---|---|---|
| 26.i. | What is Density of states? Derive an expression for density of states of a bulk semiconducting material with neat diagram. | 10 | 4 | 1 | 1 |
| ii. | Illustrate the variation of Fermi-level with temperature of N-type semiconductor. | 5 | 3 | 2 | 1 |
| 27. | Explain working and construction of a photovoltaic cell. Describe in detail solar cell characteristics curve also relate the below terms used to measure the efficiency of solar cells.
i. Short circuit current
ii. Open circuit voltage
iii. Fill factor
iv. Power maximum. | 15 | 3 | 3 | 1 |

* * * * *

THE HELPERS